THE SPIKED HARMONIC OSCILLATOR $V(r) = r^2 + \lambda r^{-4}$ AS A CHALLENGE TO PERTURBATION THEORY*)

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The standard weak- and strong-coupling perturbation series are interpreted as extreme special cases of expansions obtainable within the framework of Rayleigh-Schroedinger perturbation theory with non-diagonal propagators and unspecified zero-order energies. The formalism of the latter type is then tested by our strongly singular example. It proves suitable for applications in the domain of virtually arbitrary couplings. A few related technicalities and especially the quadruple problem of convergence are also discussed.

1. Introduction

The "spiked" central potential

$$V(\mathbf{r}) = |\mathbf{r}|^2 + \lambda|\mathbf{r}|^{-4}, \quad \lambda > 0 \tag{1}$$

is just a superposition of two exactly solvable forces [1]. As a phenomenologically appealing combination of the oscillator long-range attraction with a strongly singular short-range repulsion, it might find numerous applications in atomic, molecular or nuclear physics. As we shall see shortly the solution of the underlying Schroedinger equation

$$-\Delta\psi(\mathbf{r}) + V(\mathbf{r})\,\psi(\mathbf{r}) = E\,\psi(\mathbf{r}), \quad \|\psi\| < \infty \tag{2}$$

is not simple, especially in the small-coupling regime. In fact, it is this difficulty that makes it appealing for our purposes.

In Section 2 we present results from various non-perturbative approaches which enjoy varying degrees of success. In Section 3 we examine results from the standard weak- and strong-coupling perturbation ground-state energy expansions. This is followed in Section 4 by a brief presentation of the modified Rayleigh-Schroedinger perturbation theory (MRSPT) formalism. In an earlier work, this was shown capable of producing convergent results for the quartic anharmonic oscillator where the weak-coupling expansion is known to diverge [2, 3]. After choosing a suitable basis, the MRSPT is then applied to the present problem in Section 5. In particular, the extra freedom which is allowed will be utilized to choose as starting points (or

*) Dedicated to the memory of M. Gmitro.

zero-order approximations) the afore-mentioned weak- and strong-coupling results. Our hope and expectation is that the MRSPT will provide a bridge connecting the two domains of applicability. Finally, in Section 6 we discuss the results, as well as possible improvements and extensions.

2. Non-perturbative approaches

Perhaps the most straightforward way of solving Schroedinger equation (2) is a simple numerical integration. Choosing the boundary conditions $\psi(0) = 0 = \psi(R_{max})$, we produce successfully better approximations to the exact ground-state energy by choosing a finer and finer grid. The continuous limit is then approximated via Richardson extrapolation. We have chosen a value of 10 for the parameter R_{max}, and the results are presented in Table 1, and as advertised above, we note a steady degradation of the accuracy as λ approaches zero.

Table 1. Ground-state energies computed by an N-point integration. The row labelled "∞" is the Richardson extrapolation.

N	λ				
	0·0001	0·001	0·04	0·4	4
100	2·0995	2·99	3·372	4·025	5·554
200	2·8901	3·064	3·3823	4·030	5·55785
400	3·0130	3·066	3·384370	4·03156	5·558775
800	3·0213	3·0679	3·384908	4·03187	5·559006
1600	3·0216	3·06858	3·385043	4·031946	5·559064
3200	3·0221	3·06872	3·385077	4·031965	5·559078
∞	3·0223	3·06876	3·385088	4·031971	5·559083

The Hill-determinant technique [4] is a more ambitious algorithm of this type. It replaces the recurrent numerical generation of wavefunctions by its Laurent-series *Ansatz* representation

$$\begin{aligned} \psi(r) &= \chi(r)\left(\ldots + h_{-1}r^{-1} + h_0 + h_1 r + \ldots\right), \\ \chi(r) &= \exp\left(-\gamma r^{-1} - \kappa \ln r - \alpha r - \beta r^2/2\right), \end{aligned} \tag{3}$$

where the h's may be determined in an algebraic manner.

A common difficulty with both of these methods lies in the necessity of matching wavefunctions to the appropriate Dirichlet boundary conditions, or more generally to the WKB approximants [4]. Indeed, in the l-th partial wave, we have $\psi \approx \chi$ with, say,

$$\beta = 0\,, \quad \alpha = l(l+1)/(2\gamma)\,, \quad \kappa = -1\,, \quad \gamma = \sqrt{(\lambda)} > 0\,, \tag{4}$$

at $r \approx r_0 \ll 1$, and

$$\beta = 1\,, \quad \alpha = 0\,, \quad \kappa = \tfrac{1}{2}(1 - E)\,, \quad \gamma = 0 \tag{5}$$

at $r \approx r_\infty \gg 1$. The general $\psi(r)$ may be matched to the physical WKB estimates at an appropriately chosen pair of points (r_0, r_∞). The problem is that this WKB limiting behavior is not consistent. One inconsistency is sidestepped by choosing the zeroth partial wave, thus allowing us to set $\alpha = 0$. We are more or less forced to complete our quasi-WKB prescription by selecting the β-value of 1, and choosing κ self-consistently as described in Ref. [4]. Results are shown in Table 2.

Table 2. Ground-state energies obtained via the Hill-determinant method with the different matching points r_0 and r_∞.

λ	r_0	r_∞	Im κ	E
0·4	0·045	3·5	0·60611	4·03219
	0·040	4·0	0·60608	4·03198
0·2	0·045	3·5	0·33073	3·77306
	0·040	4·0	0·33077	3·77350

The most generally applied technique for obtaining the energy spectrum of such problems is the variational method with, say, harmonic oscillator basis, $\{|n, \omega\rangle\}$, where ω is the spring constant. Unfortunately, for the Hamiltonian of Eq. (2), this would lead to infinity. Thus, we try a modified basis of the form

$$|n, \omega, q\rangle = r^q|n, \omega\rangle\,, \tag{6}$$

which will produce matrix elements which are finite for all $q > \frac{1}{2}$, and which can be determined via compact formulae [5]. Restricting q to even integers will produce banded matrices for both the Hamiltonian and the norm, making numerical calculations and analytic analysis much easier. From now on, we shall restrict our attention to the choice $q = 2$. In Table 3 we present results for various values of λ and over

Table 3. The M_{trunc}-dependent "exact" ground-state energies generated by an M_{trunc}-dimensional diagonalization in the basis of Eq. (6) with $\omega = 1$.

M_{trunc}	λ			
	0·001	0·04	0·4	4
20	3·41	3·50	4·039	5·5596
40	3·29	3·42	4·038	5·55925
100	3·19	3·3898	4·0331	5·559085
200	3·14	3·38639	4·032017	5·559083
300	3·117	3·38635	4·031985	5·559083

a range of truncation value M_{trunc} with the parameter ω set to its "default" value of 1. Overall, the convergence is not very encouraging. However, comparison with Table 4, where the parameter ω has been optimally determined, shows a marked

Table 4. The optimized *spring constants* ω and the improved $M_{\text{trunc}} \to \infty$ convergence of the "exact" ground-state energies.

M_{trunc}	λ = 0·04		0·1		0·4		4	
	ω	$E(\omega)$	ω	$E(\omega)$	ω	$E(\omega)$	ω	$E(\omega)$
20	6·5	3·38778036	5·0	3·57789726	5·5	4·03263918	4·0	5·55910004
40	11·5	3·38616830	10·0	3·57588783	7·0	4·03198489	6·0	5·55908295
100	21·0	3·38509583	18·0	3·57555545	13·5	4·03197147	10·0	5·55908287
200	35·0	3·38508828	27·0	3·57555200	21·0	4·03197144	18·0	5·55908287

improvement, though the results still worsen as λ decreases. This is undoubtedly due to the incorrect small-r behavior of the basis which should have the asymptotic behavior of the WKB solution of Eqs. (3) and (4).

3. Standard perturbation theory: weak- and strong-coupling limits

We are fortunate to have available analytic expressions for both the weak and strong-coupling perturbative expansion for the ground state energy for our chosen Hamiltonian. The former, due to Harrell [6] and valid for small λ, is given by

$$E(\lambda) = E_0 + E_1\lambda^{1/2} + E_2\lambda + O(\lambda^{3/2})\,, \tag{7}$$

with the first two terms equal to

$$E_0 = 3\,, \quad E_1 = 4/\pi^{1/2}\,. \tag{8}$$

Computationally speaking, the use of this expansion could provide an effective method for small λ values, but the small number of known coefficients and the unknown convergence properties severely limit its usefulness.

At the other extreme, we have a strong-coupling expansion based on the fact that for large λ values the potential of Eq. (1) can be approximated to leading order by a harmonic oscillator well centered at the potential minimum. The resulting expression for the energy is given as a function of the parameter $\mu = 1/(2\lambda)^{1/6}$ by [7, 8]

$$E(\mu) = \sum_{n=-1}^{\infty} E_{2n}\mu^{2n} \tag{9}$$

with the first three terms equal to

$$E_{-2} = \tfrac{3}{2}, \quad E_0 = 6^{1/2}, \quad E_2 = \tfrac{5}{18}, \tag{10}$$

and numerical values for the next nine coefficients are found in Table 5.

An inspection of Table 5 indicates that the series (9) diverges, at least for some range of λ. In such a case, Padé approximants presumably help. Nevertheless, a comparison of Eqs. (7) and (9) shows that the expansion parameter λ enters the two formulas with different (and non-integer) exponents. Hence, any explicit interpolative connection between the strong- and weak-coupling regimes, assuming that such a connection exists, will almost certainly be quite complicated. In this paper, we intend to search for such a connection, but in a more implicit manner.

Table 5. The first ten nonzero coefficients in the strong-coupling perturbation series of Eq. (9).

n	E_n
4	−0·1464779746
6	0·0651970450
8	0·0045630050
10	−0·0821763439
12	0·1595873836
14	−0·1439179072
16	−0·2858108802
18	1·8185775355
20	−4·6069301353

4. Modified Rayleigh-Schroedinger perturbation theory: access to the intermediate coupling regime

As mentioned above, the strong-coupling expansion is almost certainly divergent, whereas the properties of the weak-coupling expansion are unknown. These troubles are in fact hardly unexpected. In previous articles [2, 3], the case of the quartic anharmonic oscillator was examined. It is generally accepted that the well-known zero radius of convergence for the ground-state energy perturbation expansion is due to the singular nature of the weak-coupling limit, where the potential suddenly changes from the quartic to the quadratic order. A modified version of Rayleigh-Schroedinger perturbation theory (MRSPT) was applied to this system, and extensive numerical results were presented with the general conclusion that the MRSPT is capable of producing a convergent energy expansion *for arbitrary coupling strengths*. This previous work leads us to the hope that a similar success may be achieved with the potential of Eq. (1).

The basic idea behind the MRSPT is a removal of the usual constraint that the unperturbed Hamiltonian H_0 be diagonal in the chosen basis. This is accomplished by a repartition of the Hamiltonian

$$H(\lambda) = H_0 + \varrho W(\lambda) \tag{11}$$

via the addition and subtraction of the term $\varrho P\xi P$,

$$H_0 = \tilde{H}_0 + \varrho P\xi P\,, \tag{12}$$

$$\varrho W = H(\lambda) - \tilde{H}_0 - \varrho P\xi P\,. \tag{13}$$

The above equations need some explanation but for full details the reader must consult Ref. [3]. We use λ to represent the coupling constant(s), and ϱ is a formal expansion parameter. Using the notation of, for example, Löwdin et al. [9], we have partitioned our chosen full space (after a truncation to M_{cut} states) into a model space P of dimension t and an out-of-model space Q of dimension $M_{\mathrm{cut}} - t$. The repartition, produced by the addition and subtraction of the in-model-space correction $\varrho P\xi P$ is used to render the solution of the in-model-space projection of the t zeroth-order Schroedinger equations, $P(H_0 - \mathscr{E})\,|\psi\rangle = 0$, trivial. This is accomplished by choosing the specific form of the auxiliary matrix,

$$\xi = \varrho^{-1}P(\mathscr{E} - \tilde{H}_0 - \tilde{H}_0 QRQ\tilde{H}_0)\,P\,. \tag{14}$$

Two more comments will complete our very brief description. First, we have imposed a t-fold degeneracy within the model space, which in turn implies that we need only a single input resolvent $R = (\mathscr{E} - QH_0Q)^{-1}$ in order to carry out our prescription.

In summary, a specific repartition of our Hamiltonian produces a perturbation expansion which allows one to choose as the zeroth-order approximation an *arbitrary* Hamiltonian. Within the model space, the resultant set of t Schroedinger equations is made t-fold degenerate, and will have as solution the *arbitrary parameter* $\mathscr{E}$. We thus have at our disposal the free parameter $\mathscr{E}$, only indirectly associated with the zeroth-order H_0 [2].

One should expect that the essentially arbitrary choice of H_0 may allow us to generate an expansion for the energy which behaves much better and is rapidly convergent. Of course, there is a price to be paid, namely that one must calculate the resolvent R. Various algebraic techniques have been used to do this efficiently in the case (typical so far) where H_0 is banded. In particular, continued fractions and their generalizations can be used here to reduce this to a problem whose complexity grows linearly with the basis size [10]. Additionally, one may use the fixed-point techniques to extrapolate to the infinite limit [11].

5. MRSPT: Numerical results

We shall discuss here briefly, to avoid confusion latter on, the convergence problem. We have inherent in the MRSPT four distinct truncations to make. Let us assume that we have in general an infinite set of basis states. First, the zeroth-order Hamiltonian H_0 and the corresponding resolvent R will be calculated in a truncated basis of size M_{full}. Next, we may truncate ϱW to a size M_{cut} before carrying out the perturbation expansion. Thirdly, we must fix the perturbation order N and, finally, we can consider convergence as the model space size t increases.

All experience has shown that the first convergence problem, involving M_{full}, is not really a problem. The second convergence, M_{cut}, is clearly very much model- and basis-specific. In this paper, we shall choose to implement the MRSPT within the variationally-optimized basis of Eq. (6). As we are mainly interested in analyzing the properties of the MRSPT expansion itself, we really need only to concentrate on the final two convergence problems once the basis size has been fixed. Thus, once we have selected a basis and a value of $M_{\text{full}} = M_{\text{cut}}$, the energy value produced here via diagonalization shall be our "exact" answer. Convergence to this answer will indicate *full* convergence of the MRS expansion within the chosen truncated basis, and we cannot expect to obtain more than this. Our main interest lies in examining the effect of the free parameter $\mathscr{E}$ (the zeroth-order starting point) on the final results.

Table 6. Recurrent determination of the MRSPT starting parameter $\mathscr{E}$, with $M_{\text{cut}} = 20$, $\lambda = 0{\cdot}4$, $\lambda_0 = 0{\cdot}38$, and with ω set equal to the optimal value 5·5 (cf. Table 4).

Order	$\mathscr{E}$
0	3·00000000
1	4·23401239
2	4·04379384
3	4·03275837
4	4·03277019
5	4·03277015
6	4·03277015

We have used a method for determining the free parameter $\mathscr{E}$, which is illustrated in Table 6. These results are for $M_{\text{full}} = 20$, $\lambda = 0{\cdot}4$, and with λ_0 chosen as above. Taking as (a very poor) starting point the value of 3, we define $\mathscr{E}$ recursively as the first-order MRSPT energy until convergence is reached. This took 5 iterations in our table, and the final converged $\mathscr{E}$ value produces an expansion which is even slightly better than that produced by the "optimal" choice, with $\xi = 0$ in Eq. (14).

The latter observation is not so surprising – our algorithm simply takes more contributions into account immediately. Second interesting point seems more puzzling: The recurrence procedure for $\mathscr{E}$, while convergent, does not in fact converge to the exact energy, i.e., the higher-order corrections never become negligible for $\lambda_0 \neq \lambda$. This numerical curiosity remains so far unexplained.

In Table 7, we present a variety of results for the ground-state energy calculated via the MRSPT. We have used a basis size M_{cut} of 100, and have set $\lambda_0 = 0{\cdot}95\lambda$. Here and below we always use an initial basis size of $M_{\text{full}} = M_{\text{cut}}$. The first two sets of data have as starting points $\mathscr{E}$ the low-order weak- and strong-coupling approximations given by Eqs. (8) and (10), respectively. It is clear that the range

Table 7. MRSPT results with input $\mathscr{E}$ given by its weak- and strong-coupling estimates and by its selfconsistent $\xi = 0$ value (optimal ω, $M_{\text{cut}} = 100$ and $\lambda_0 = 0{\cdot}95\lambda$).

$\mathscr{E}$	Order	λ			
		0·04	0·1	0·4	4
WC, Eq. (7)	0	3·451	3·696	4·43	7·5
	10	3·3850958	3·5755554	4·031964	−241·0
SC, Eq. (9)	0	3·741	3·795	4·123	5·540
	10	3·3850950	3·5755554	4·0319715	5·5590829
$\xi(\mathscr{E}) = 0$	0	3·376	3·563	4·010	5·510
	10	3·3850958	3·5755554	4·0319715	5·5590829

of applicability of both expansions has been greatly extended. The next set of data, labeled $\xi = 0$, has the zeroth-order $\mathscr{E}$ equal to the exact ground-state energy for the M_{cut} basis for the given value of λ_0. As in previous work with other potentials [3], we see that once again we obtain complete convergence *independent of* λ.

More importantly, there is some range of λ (demonstrated here at the value of 0·1) where the MRS expansions generated from both the strong- and weak-coupling starting points become convergent. It should be remembered that the original strong-coupling expansion at such a small λ value is *very* divergent.

Table 8. The MRSPT low-lying spectrum in the growing model space, with $M_{\text{full}} = M_{\text{cut}} = 20$, $\lambda = 0{\cdot}4$, $\lambda_0 = 0{\cdot}38$ and optimal $\omega = 5{\cdot}5$.

Energy level	Pert. order	Dimension t				
		1	2	4	6	8
0	1	4·03298	4·032718	4·032671	4·0326557	4·0326490
	2	4·03262958	4·03263751	4·03263840	4·03263874	4·03263891
1	1		13·976	9·846	8·855	8·516
	2		9·6817	8·8496	8·5170	8·3904
2	1			22·87	17·19	14·84
	2			17·930	15·169	13·860

In Table 8 we present results for the ground and first few excited states calculated with varying model space size, up to second order. Remember that all states have $\mathscr{E}$ as their zeroth-order approximation. As expected, the accuracy degrades with increasing excitation level. At the same time, we see that increasing the model space size t produces a rapid increase in accuracy for a given level. For example, the $t = 8$ second-order result is almost as accurate as the $t = 1$ fourth-order energy. This is an entirely unexpected phenomenon which might find applications as yet another method of improving the accuracy within the MRSPT formalism.

6. Discussion and conclusions

We may conclude that the MRSPT formalism is capable of producing highly accurate and rapidly-convergent results even for such a singular perturbation as examined in the present paper. In addition, we have also seen that the results are produced in a way which is essentially independent of the strength of the coupling constant again. Such a conclusion is highly satisfactory and re-confirms the good applicability of the perturbative formalisms very far from the diagonalized H_0's.

The calculated spectra are in fact not overly dependent on the choice of the MRSPT input or free parameter $\mathscr{E}$ itself. This is a crucial point. While the *a priori* optimal value for $\mathscr{E}$ seems to be very close to the energy of the unperturbed Hamiltonian H_0, we clearly cannot know this value beforehand, unless we return back to the standard formalisms with diagonal H_0. At the same time, the choice of $\mathscr{E}$ is not entirely arbitrary, as it specifies the energy levels and it also seems relevant to the acceleration of convergence.

A core of our present paper lies in noticing that there exist very "natural" ways of choosing the parameter $\mathscr{E}$ which do not require any knowledge of H_0. The low-order weak- and strong-coupling expansions were tested as recommended candidates – we have seen that they provide good starting values over a wide range of coupling constants. The resultant MRS expansions proved once again rapidly convergent.

In a way, the latter result combines MRSPT technique with simpler "input" perturbative estimates. In a broader methodical context, this may be interpreted as a fulfillment of the dreams of perturbation theorists: The MRS formalism proves to work as an interpolative prescription at intermediate couplings, once we have computed two independent strong- and weak-coupling estimates as input.

The MRSPT technique may also be used as a selfconsistent prescription: In place of the external estimates, we may use the zero-order MRSPT result itself. Numerically, we have shown that such a simple recursive prescription can produce an excellent final $\mathscr{E}$ value starting from an essentially arbitrary point. In this way, any *a priori* knowledge of the spectrum of H_0 becomes obviated altogether.

In a broader context, we now emphasize a few purely methodical general aspects of our numerically generated conclusions.

1. The failure to converge or the slowness of convergence for the type of problem we have examined here may be traced back to the inadequacy of the zeroth-order Hamiltonian. Within the MRSPT framework, we are free to separate H_0 and the unperturbed basis $|n\rangle$ – the former operator need not be diagonal in the latter basis. This is clearly a very satisfactory freedom. Indeed, the precision of perturbative results depends on the quality of the approximate Hamiltonian H_0, while a feasibility of computations often depends on the basis itself. Hence, any extension of our present work to more complex and/or realistic systems promises to be straightforward.
2. It is very important that an *a priori* knowledge of the spectrum of the zeroth-order Hamiltonian is not needed in the MRSPT formalism. We have shown that even a very rough guess of the zero-order energy $\mathscr{E}$ is needed – its approximation by the slowly convergent (here: strong- and weak-coupling) and/or low-order (here: MRSPT) prescriptions proved entirely sufficient. Thus, there exist several independent possibilities of how the sufficiently satisfactory value of the MRSPT input parameter $\mathscr{E}$ can be generated in most applications.
3. The MRS technique still leaves a lot of the formal questions unanswered. One of them concerns the distinction between the "sufficient" truncation of the basis M_{full} (needed, e.g., in the evaluations of the propagators) and the "sufficient" truncation M_{cut} pertinent to the matrix of perturbations $V = H - H_0$. We also do not know how to vary the form of H_0 and how to get an optimal non-diagonal simplification of a given realistic Hamiltonian. Basically, we have to optimize the total (in fact, quadruple) convergence of our MRSPT expansions. Mathematically, this is a problem much more difficult than its standard Rayleigh-Schroedinger counterpart.
4. Our present numerical study made use of the apparently very non-perturbative, strongly spiked oscillators [6]. We believe that our tests may also serve as a useful guide towards the more realistic computations or, complementarily, more rigorous analyses and further improvements of the formalism. In principle, such a future work should parallel the similar developments (re-summations, relations to the renormalization group, etc.) achieved already with the diagonal zero order Hamiltonians and unperturbed propagators. Of course, this is a far reaching program which already lies too far beyond the scope of the present single paper.

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